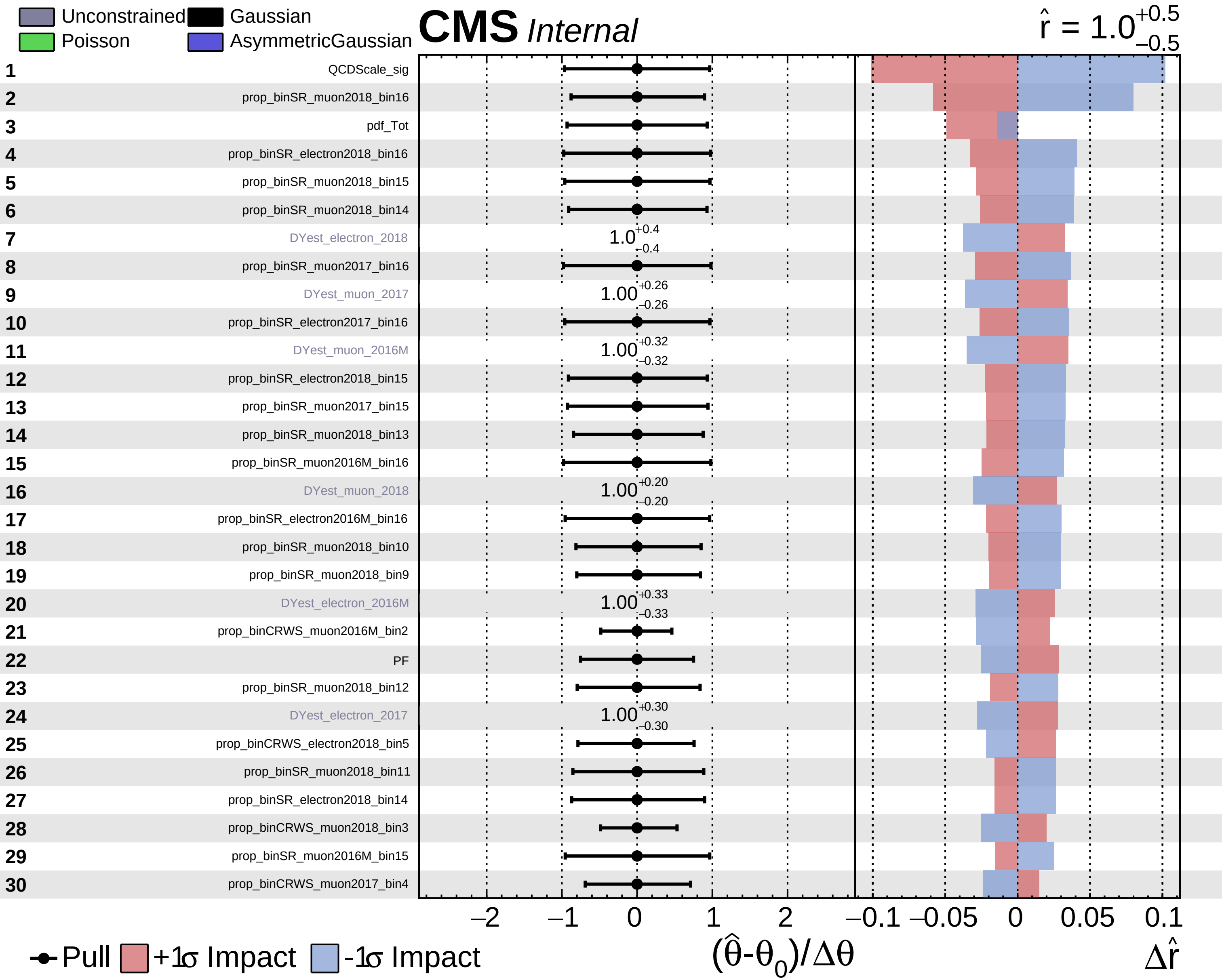
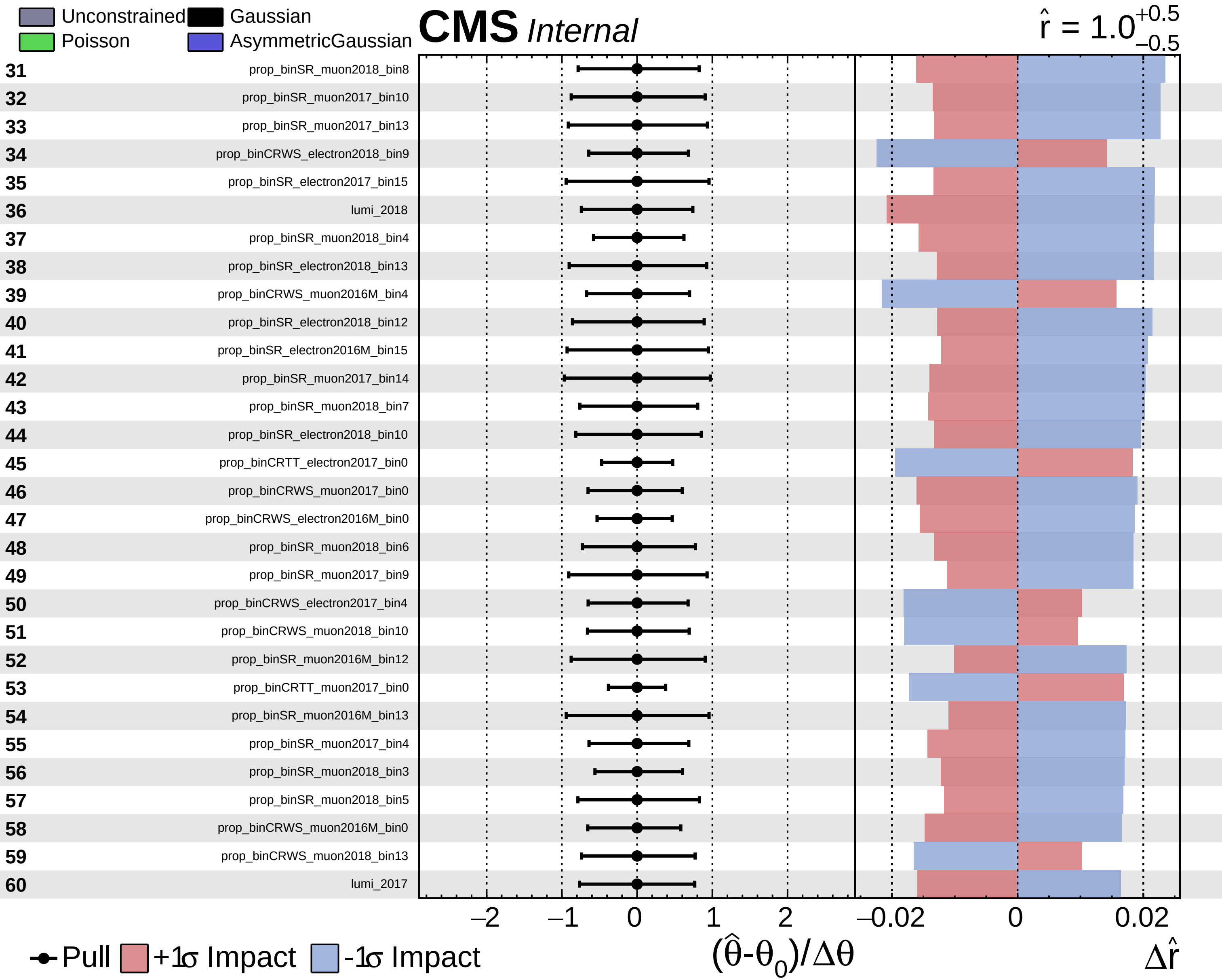
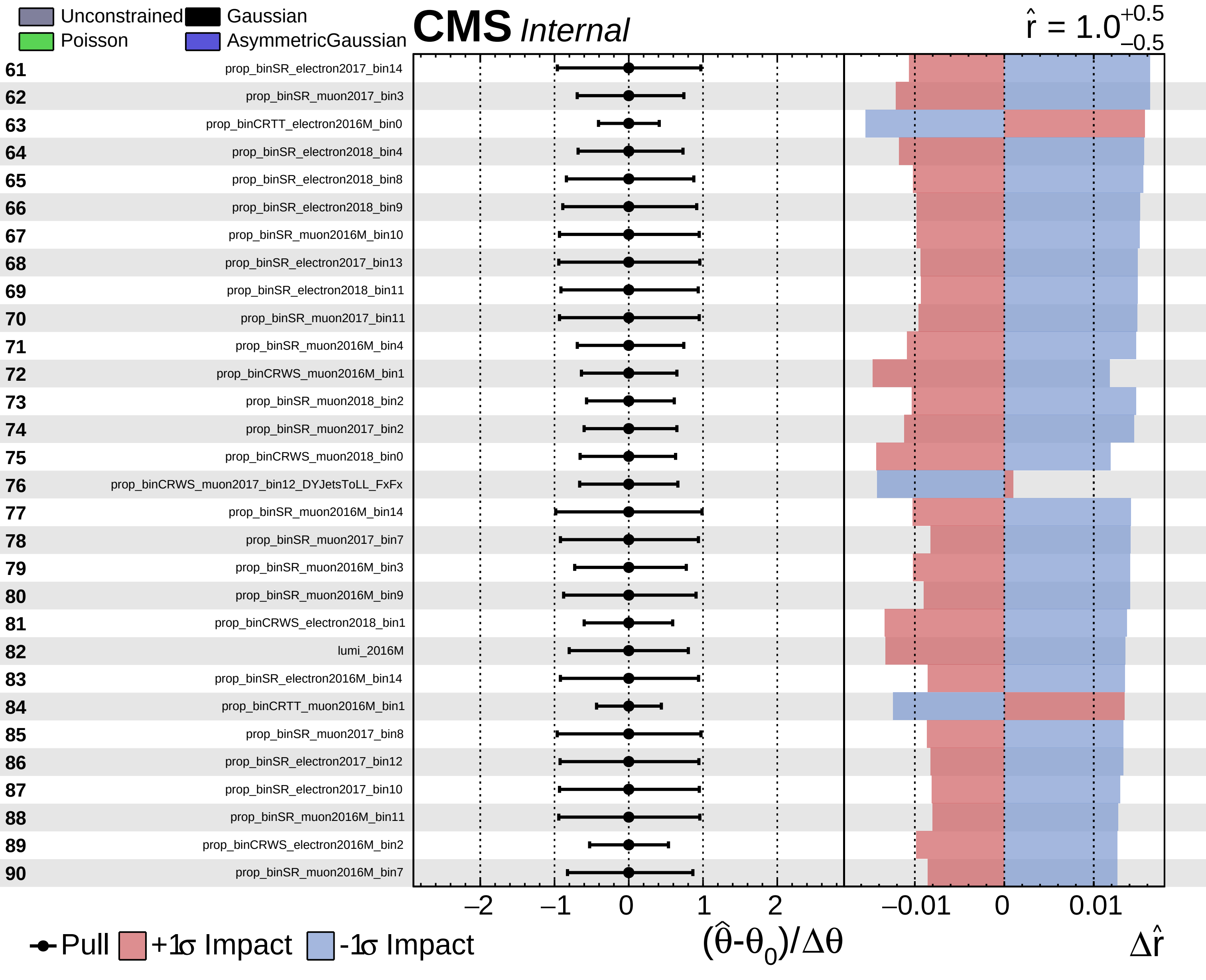
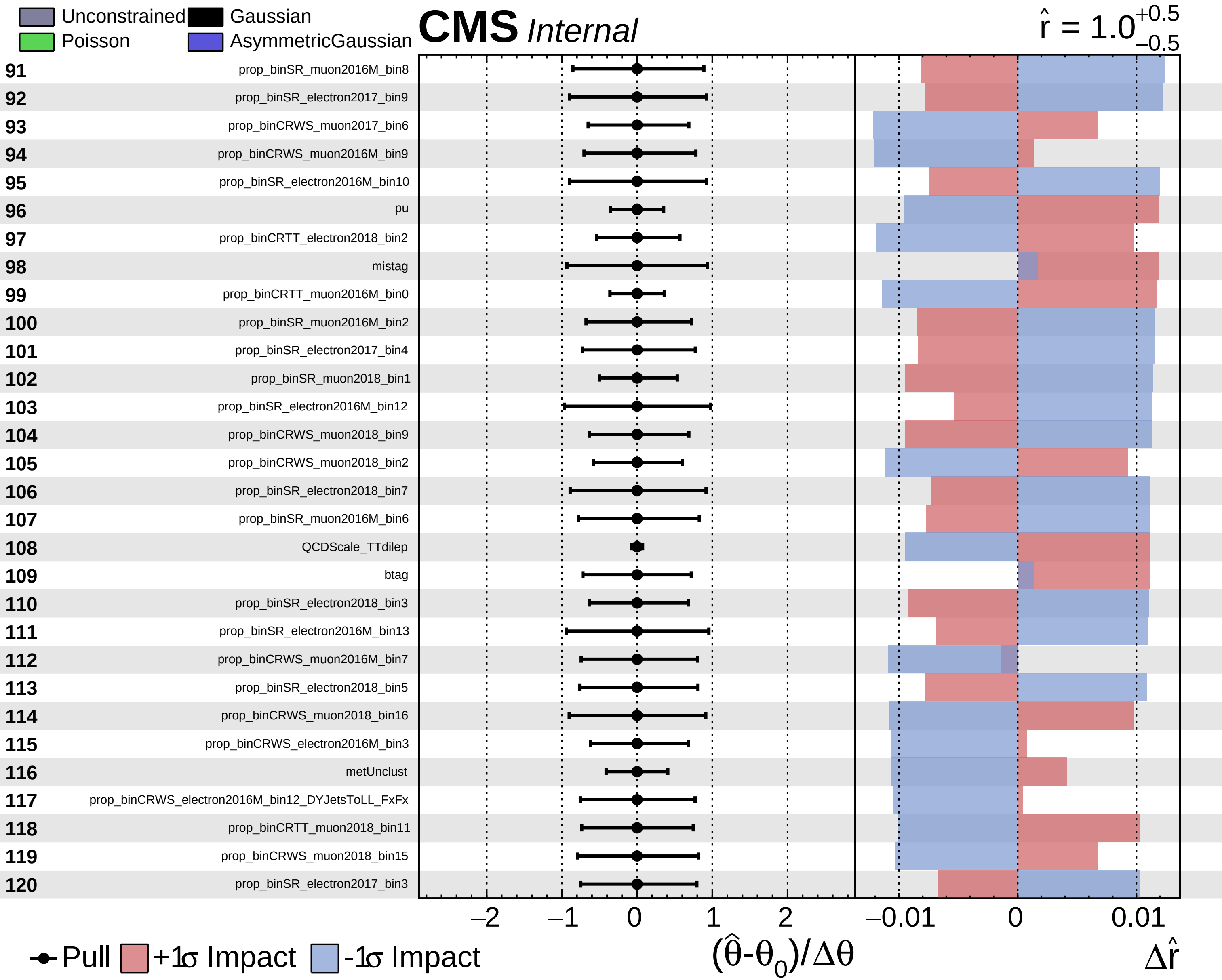


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
1 QCDScale_sig
2 prop_binSR_muon2018_bin16
3 pdf_Tot
4 prop_binSR_electron2018_bin16
5 prop_binSR_muon2018_bin15
6 prop_binSR_muon2018_bin14
7 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
8 prop_binSR_muon2017_bin16
9 DYest_muon_2017 $1.00^{+0.26}_{-0.26}$
10 prop_binSR_electron2017_bin16
11 DYest_muon_2016M $1.00^{+0.32}_{-0.32}$
12 prop_binSR_electron2018_bin15
13 prop_binSR_muon2017_bin15
14 prop_binSR_muon2018_bin13
15 prop_binSR_muon2016M_bin16
16 DYest_muon_2018 $1.00^{+0.20}_{-0.20}$
17 prop_binSR_electron2016M_bin16
18 prop_binSR_muon2018_bin10
19 prop_binSR_muon2018_bin9
20 DYest_electron_2016M $1.00^{+0.33}_{-0.33}$
21 prop_binCRWS_muon2016M_bin2
22 PF
23 prop_binSR_muon2018_bin12
24 DYest_electron_2017 $1.00^{+0.30}_{-0.30}$
25 prop_binCRWS_electron2018_bin5
26 prop_binSR_muon2018_bin11
27 prop_binSR_electron2018_bin14
28 prop_binCRWS_muon2018_bin3
29 prop_binSR_muon2016M_bin15
30 prop_binCRWS_muon2017_bin4
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

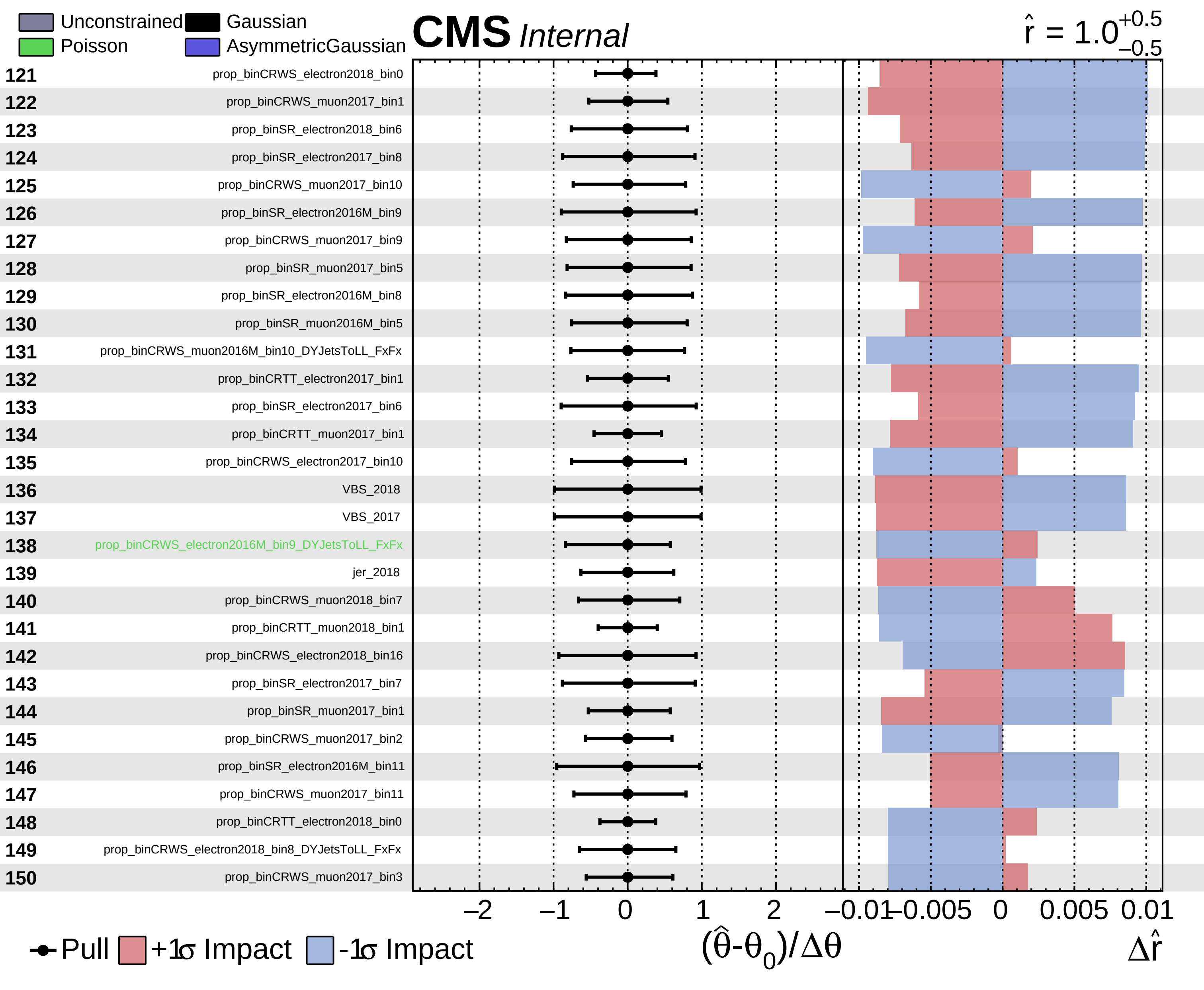

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
31 prop_binSR_muon2018_bin8
32 prop_binSR_muon2017_bin10
33 prop_binSR_muon2017_bin13
34 prop_binCRWS_electron2018_bin9
35 prop_binSR_electron2017_bin15
36 lumi_2018
37 prop_binSR_muon2018_bin4
38 prop_binSR_electron2018_bin13
39 prop_binCRWS_muon2016M_bin4
40 prop_binSR_electron2018_bin12
41 prop_binSR_electron2016M_bin15
42 prop_binSR_muon2017_bin14
43 prop_binSR_muon2018_bin7
44 prop_binSR_electron2018_bin10
45 prop_binCRTT_electron2017_bin0
46 prop_binCRWS_muon2017_bin0
47 prop_binCRWS_electron2016M_bin0
48 prop_binSR_muon2018_bin6
49 prop_binSR_muon2017_bin9
50 prop_binCRWS_electron2017_bin4
51 prop_binCRWS_muon2018_bin10
52 prop_binSR_muon2016M_bin12
53 prop_binCRTT_muon2017_bin0
54 prop_binSR_muon2016M_bin13
55 prop_binSR_muon2017_bin4
56 prop_binSR_muon2018_bin3
57 prop_binSR_muon2018_bin5
58 prop_binCRWS_muon2016M_bin0
59 prop_binCRWS_muon2018_bin13
60 lumi_2017
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



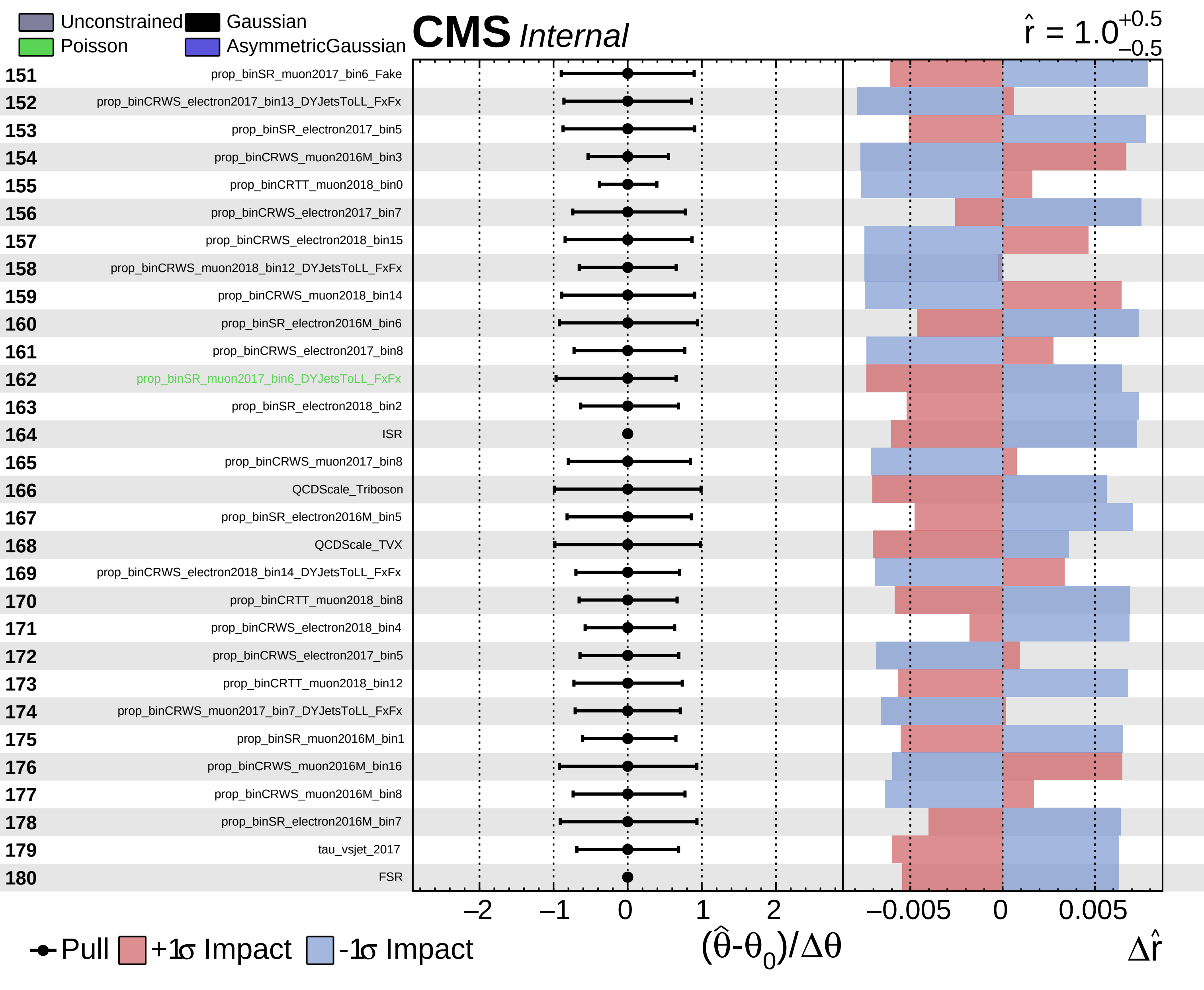
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
61 prop_binSR_electron2017_bin14
62 prop_binSR_muon2017_bin3
63 prop_binCRTT_electron2016M_bin0
64 prop_binSR_electron2018_bin4
65 prop_binSR_electron2018_bin8
66 prop_binSR_electron2018_bin9
67 prop_binSR_muon2016M_bin10
68 prop_binSR_electron2017_bin13
69 prop_binSR_electron2018_bin11
70 prop_binSR_muon2017_bin11
71 prop_binSR_muon2016M_bin4
72 prop_binCRWS_muon2016M_bin1
73 prop_binSR_muon2018_bin2
74 prop_binSR_muon2017_bin2
75 prop_binCRWS_muon2018_bin0
76 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
77 prop_binSR_muon2016M_bin14
78 prop_binSR_muon2017_bin7
79 prop_binSR_muon2016M_bin3
80 prop_binSR_muon2016M_bin9
81 prop_binCRWS_electron2018_bin1
82 lumi_2016M
83 prop_binSR_electron2016M_bin14
84 prop_binCRTT_muon2016M_bin1
85 prop_binSR_muon2017_bin8
86 prop_binSR_electron2017_bin12
87 prop_binSR_electron2017_bin10
88 prop_binSR_muon2016M_bin11
89 prop_binCRWS_electron2016M_bin2
90 prop_binSR_muon2016M_bin7
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

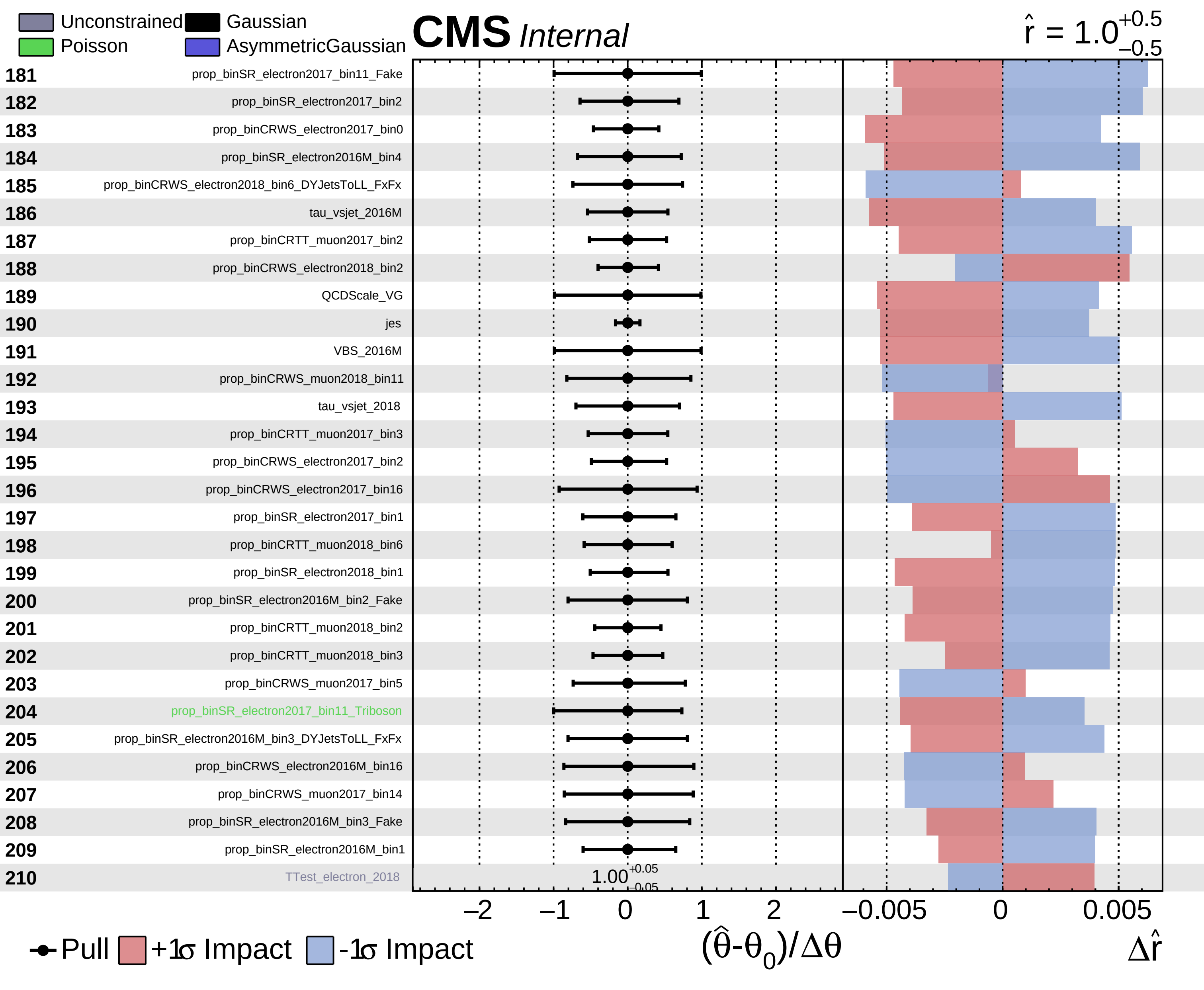


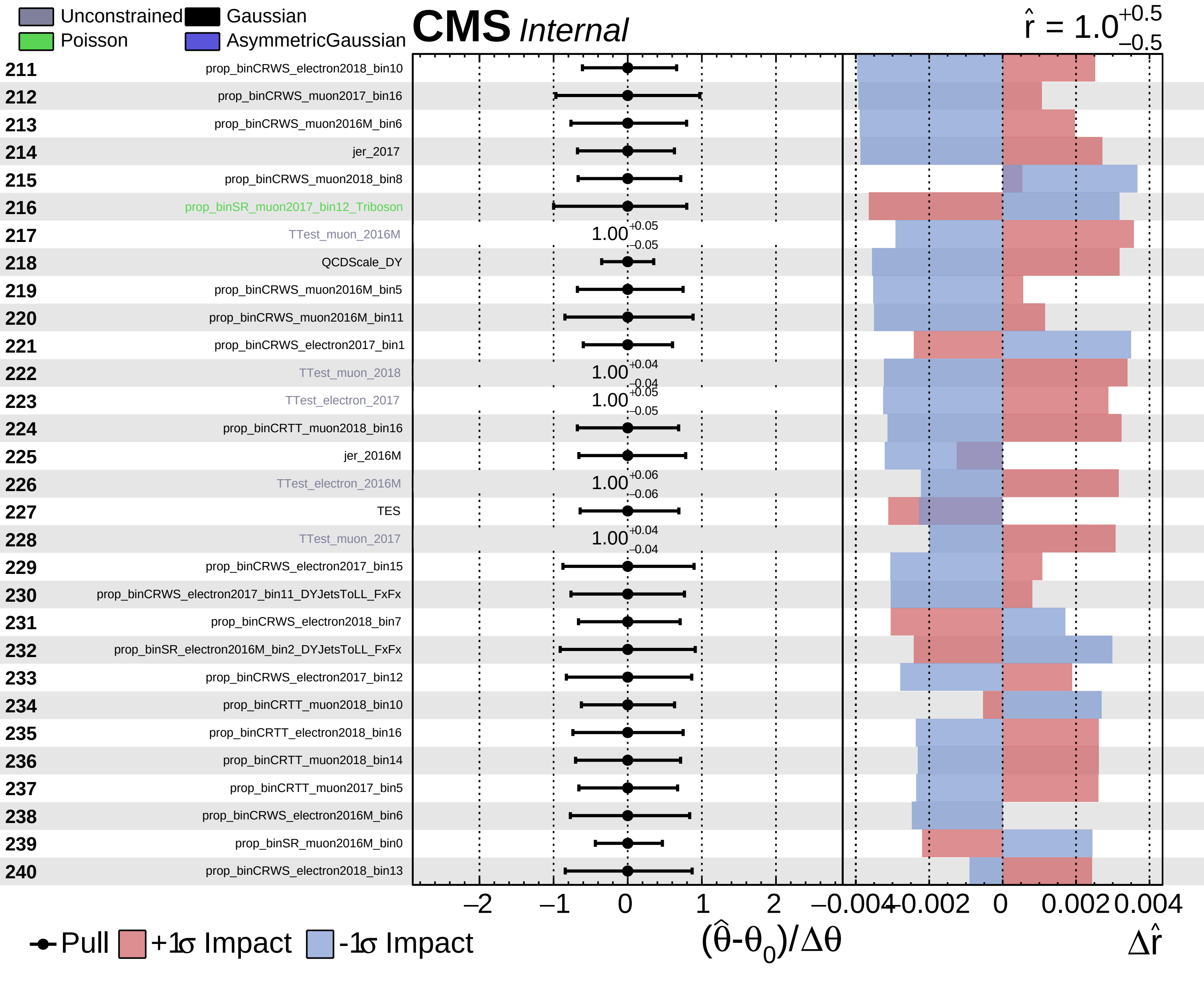
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
91 prop_binSR_muon2016M_bin8
92 prop_binSR_electron2017_bin9
93 prop_binCRWS_muon2017_bin6
94 prop_binCRWS_muon2016M_bin9
95 prop_binSR_electron2016M_bin10
96 pu
97 prop_binCRTT_electron2018_bin2
98 mistag
99 prop_binCRTT_muon2016M_bin0
100 prop_binSR_muon2016M_bin2
101 prop_binSR_electron2017_bin4
102 prop_binSR_muon2018_bin1
103 prop_binSR_electron2016M_bin12
104 prop_binCRWS_muon2018_bin9
105 prop_binCRWS_muon2018_bin2
106 prop_binSR_electron2018_bin7
107 prop_binSR_muon2016M_bin6
108 QCDScale_TTdilep
109 btag
110 prop_binSR_electron2018_bin3
111 prop_binSR_electron2016M_bin13
112 prop_binCRWS_muon2016M_bin7
113 prop_binSR_electron2018_bin5
114 prop_binCRWS_muon2018_bin16
115 prop_binCRWS_electron2016M_bin3
116 metUnclust
117 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
118 prop_binCRTT_muon2018_bin11
119 prop_binCRWS_muon2018_bin15
120 prop_binSR_electron2017_bin3
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



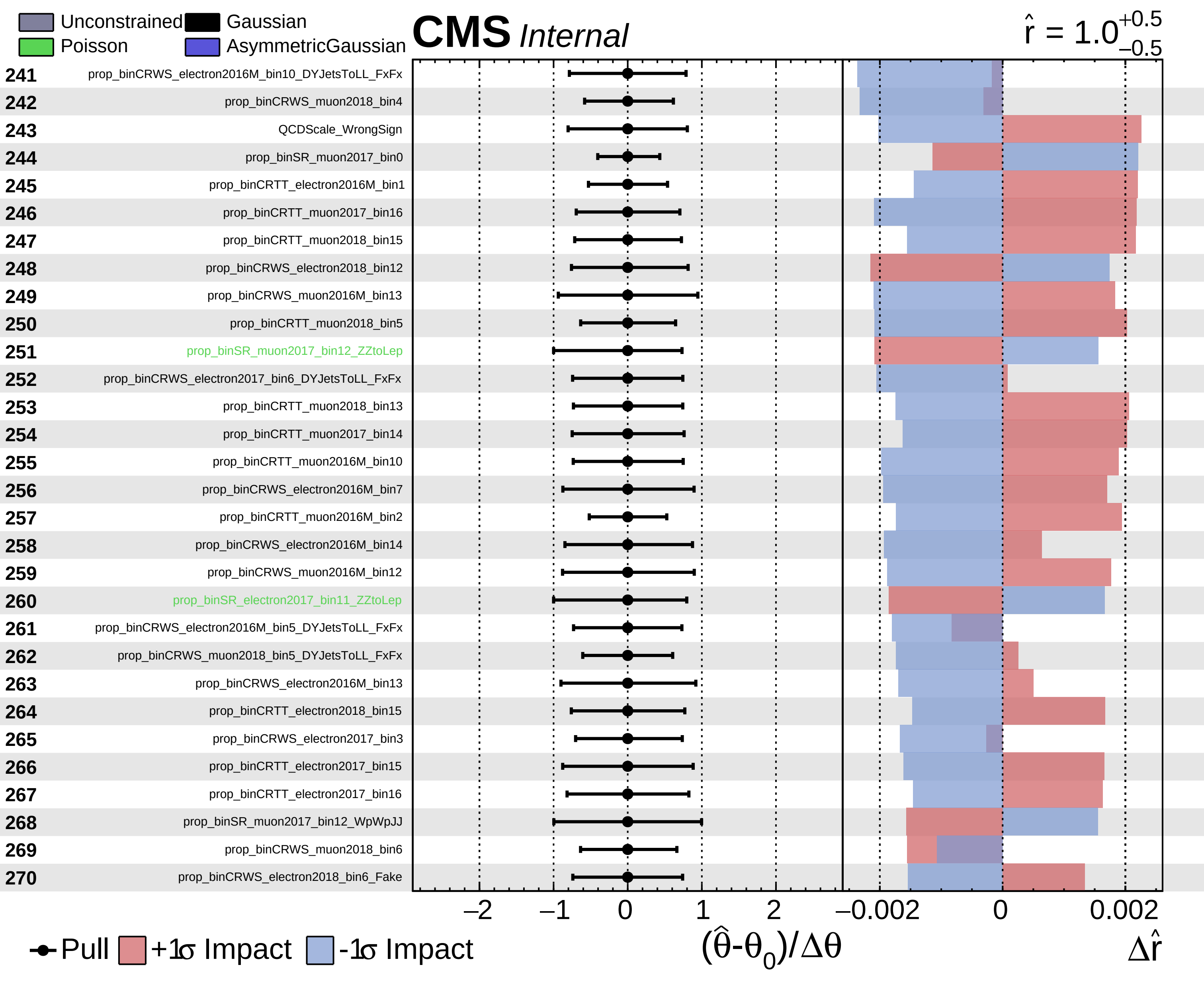
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
121 prop_binCRWS_electron2018_bin0
122 prop_binCRWS_muon2017_bin1
123 prop_binSR_electron2018_bin6
124 prop_binSR_electron2017_bin8
125 prop_binCRWS_muon2017_bin10
126 prop_binSR_electron2016M_bin9
127 prop_binCRWS_muon2017_bin9
128 prop_binSR_muon2017_bin5
129 prop_binSR_electron2016M_bin8
130 prop_binSR_muon2016M_bin5
131 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
132 prop_binCRTT_electron2017_bin1
133 prop_binSR_electron2017_bin6
134 prop_binCRTT_muon2017_bin1
135 prop_binCRWS_electron2017_bin10
136 VBS_2018
137 VBS_2017
138 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
139 jer_2018
140 prop_binCRWS_muon2018_bin7
141 prop_binCRTT_muon2018_bin1
142 prop_binCRWS_electron2018_bin16
143 prop_binSR_electron2017_bin7
144 prop_binSR_muon2017_bin1
145 prop_binCRWS_muon2017_bin2
146 prop_binSR_electron2016M_bin11
147 prop_binCRWS_muon2017_bin11
148 prop_binCRTT_electron2018_bin0
149 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
150 prop_binCRWS_muon2017_bin3
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
151 prop_binSR_muon2017_bin6_Fake
152 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
153 prop_binSR_electron2017_bin5
154 prop_binCRWS_muon2016M_bin3
155 prop_binCRTT_muon2018_bin0
156 prop_binCRWS_electron2017_bin7
157 prop_binCRWS_electron2018_bin15
158 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
159 prop_binCRWS_muon2018_bin14
160 prop_binSR_electron2016M_bin6
161 prop_binCRWS_electron2017_bin8
162 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
163 prop_binSR_electron2018_bin2
164 ISR
165 prop_binCRWS_muon2017_bin8
166 QCDScale_Triboson
167 prop_binSR_electron2016M_bin5
168 QCDScale_TVX
169 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
170 prop_binCRTT_muon2018_bin8
171 prop_binCRWS_electron2018_bin4
172 prop_binCRWS_electron2017_bin5
173 prop_binCRTT_muon2018_bin12
174 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
175 prop_binSR_muon2016M_bin1
176 prop_binCRWS_muon2016M_bin16
177 prop_binCRWS_muon2016M_bin8
178 prop_binSR_electron2016M_bin7
179 tau_vsjet_2017
180 FSR
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

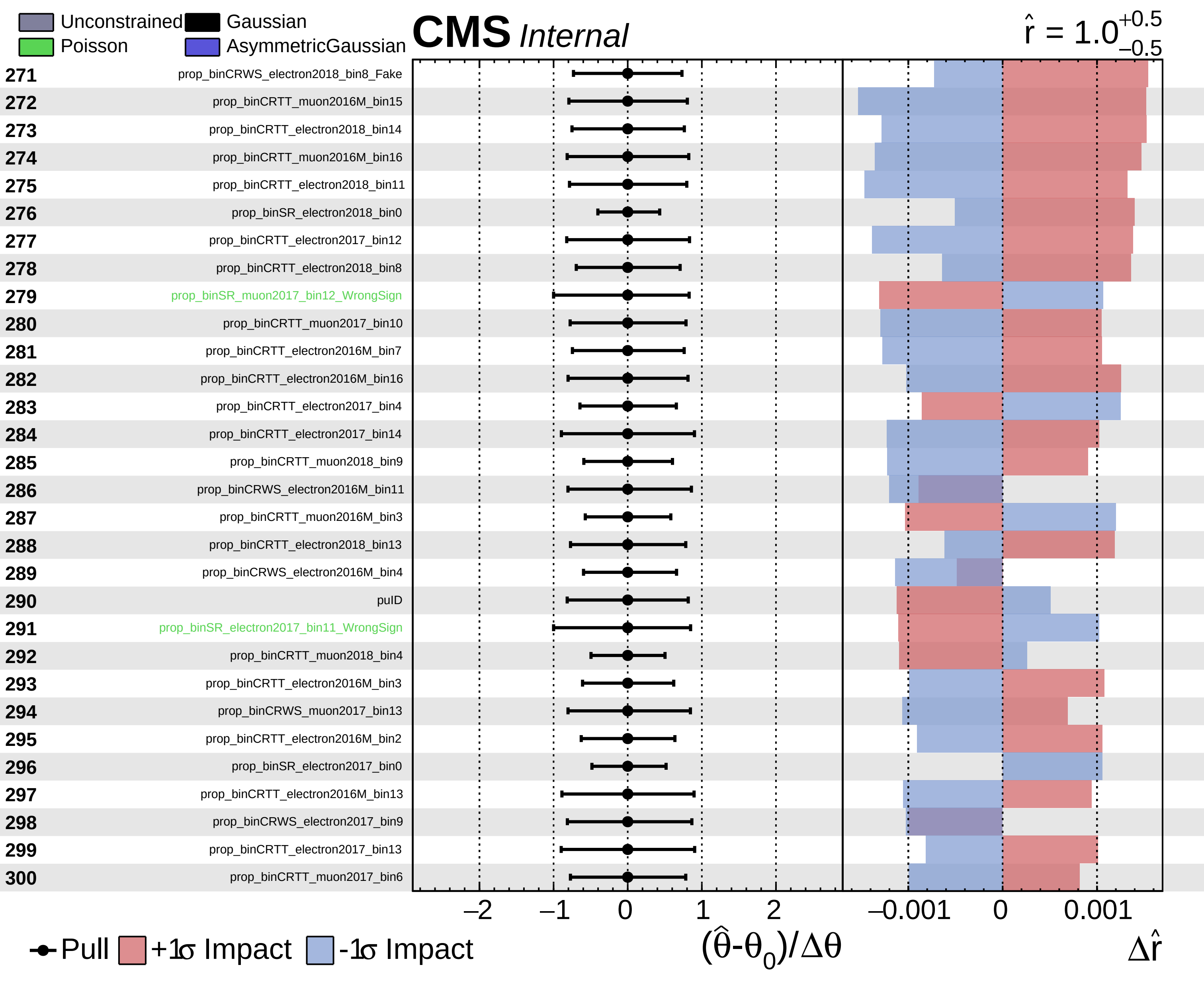

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
181 prop_binSR_electron2017_bin11_Fake
182 prop_binSR_electron2017_bin2
183 prop_binCRWS_electron2017_bin0
184 prop_binSR_electron2016M_bin4
185 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
186 tau_vsjet_2016M
187 prop_binCRTT_muon2017_bin2
188 prop_binCRWS_electron2018_bin2
189 QCDScale_VG
190 jes
191 VBS_2016M
192 prop_binCRWS_muon2018_bin11
193 tau_vsjet_2018
194 prop_binCRTT_muon2017_bin3
195 prop_binCRWS_electron2017_bin2
196 prop_binCRWS_electron2017_bin16
197 prop_binSR_electron2017_bin1
198 prop_binCRTT_muon2018_bin6
199 prop_binSR_electron2018_bin1
200 prop_binSR_electron2016M_bin2_Fake
201 prop_binCRTT_muon2018_bin2
202 prop_binCRTT_muon2018_bin3
203 prop_binCRWS_muon2017_bin5
204 prop_binSR_electron2017_bin11_Triboson
205 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
206 prop_binCRWS_electron2016M_bin16
207 prop_binCRWS_muon2017_bin14
208 prop_binSR_electron2016M_bin3_Fake
209 prop_binSR_electron2016M_bin1
210 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



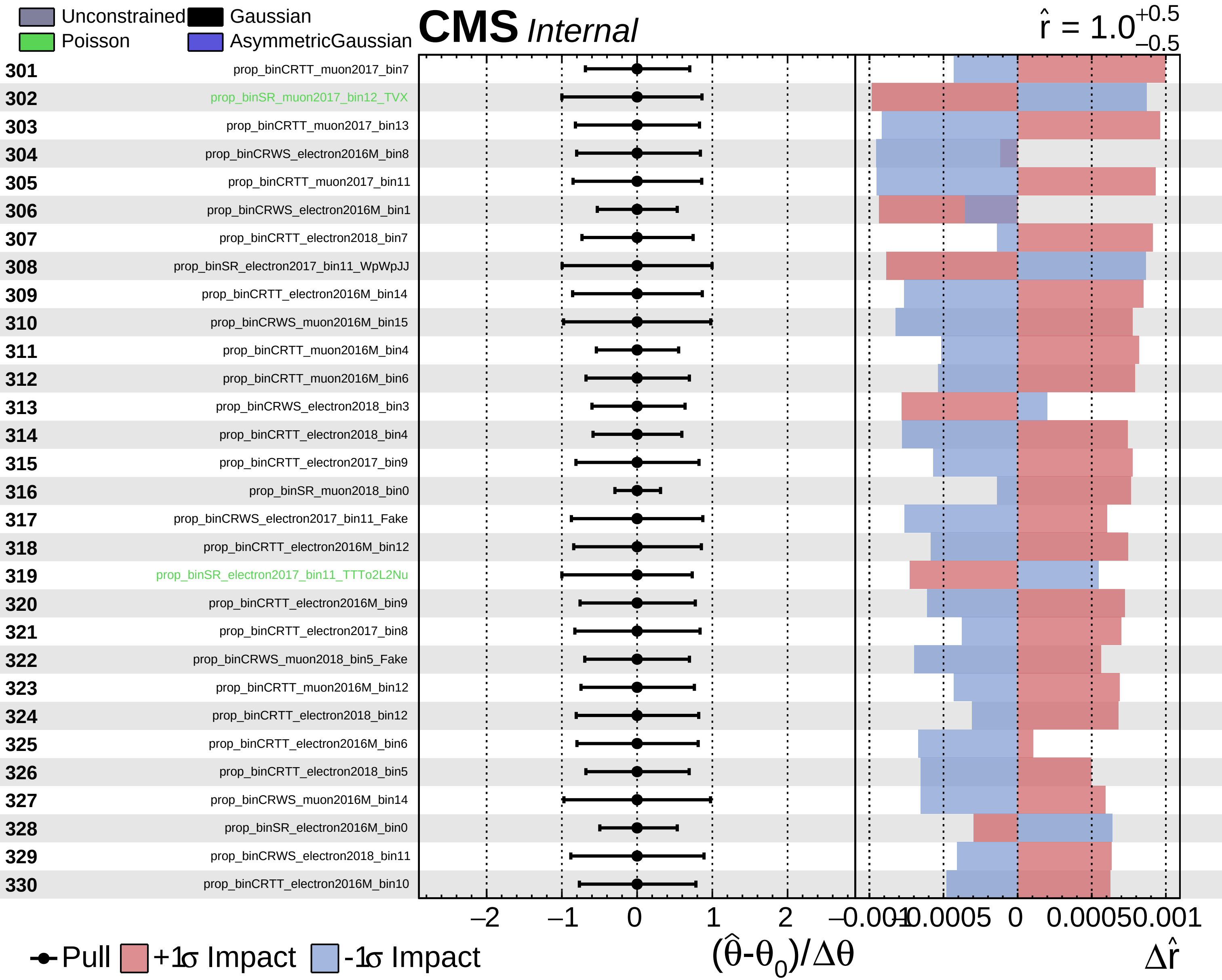
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
211 prop_binCRWS_electron2018_bin10
212 prop_binCRWS_muon2017_bin16
213 prop_binCRWS_muon2016M_bin6
214 jer_2017
215 prop_binCRWS_muon2018_bin8
216 prop_binSR_muon2017_bin12_Triboson
217 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
218 QCDScale_DY
219 prop_binCRWS_muon2016M_bin5
220 prop_binCRWS_muon2016M_bin11
221 prop_binCRWS_electron2017_bin1
222 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
223 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
224 prop_binCRTT_muon2018_bin16
225 jer_2016M
226 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
227 TES
228 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
229 prop_binCRWS_electron2017_bin15
230 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
231 prop_binCRWS_electron2018_bin7
232 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
233 prop_binCRWS_electron2017_bin12
234 prop_binCRTT_muon2018_bin10
235 prop_binCRTT_electron2018_bin16
236 prop_binCRTT_muon2018_bin14
237 prop_binCRTT_muon2017_bin5
238 prop_binCRWS_electron2016M_bin6
239 prop_binSR_muon2016M_bin0
240 prop_binCRWS_electron2018_bin13
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



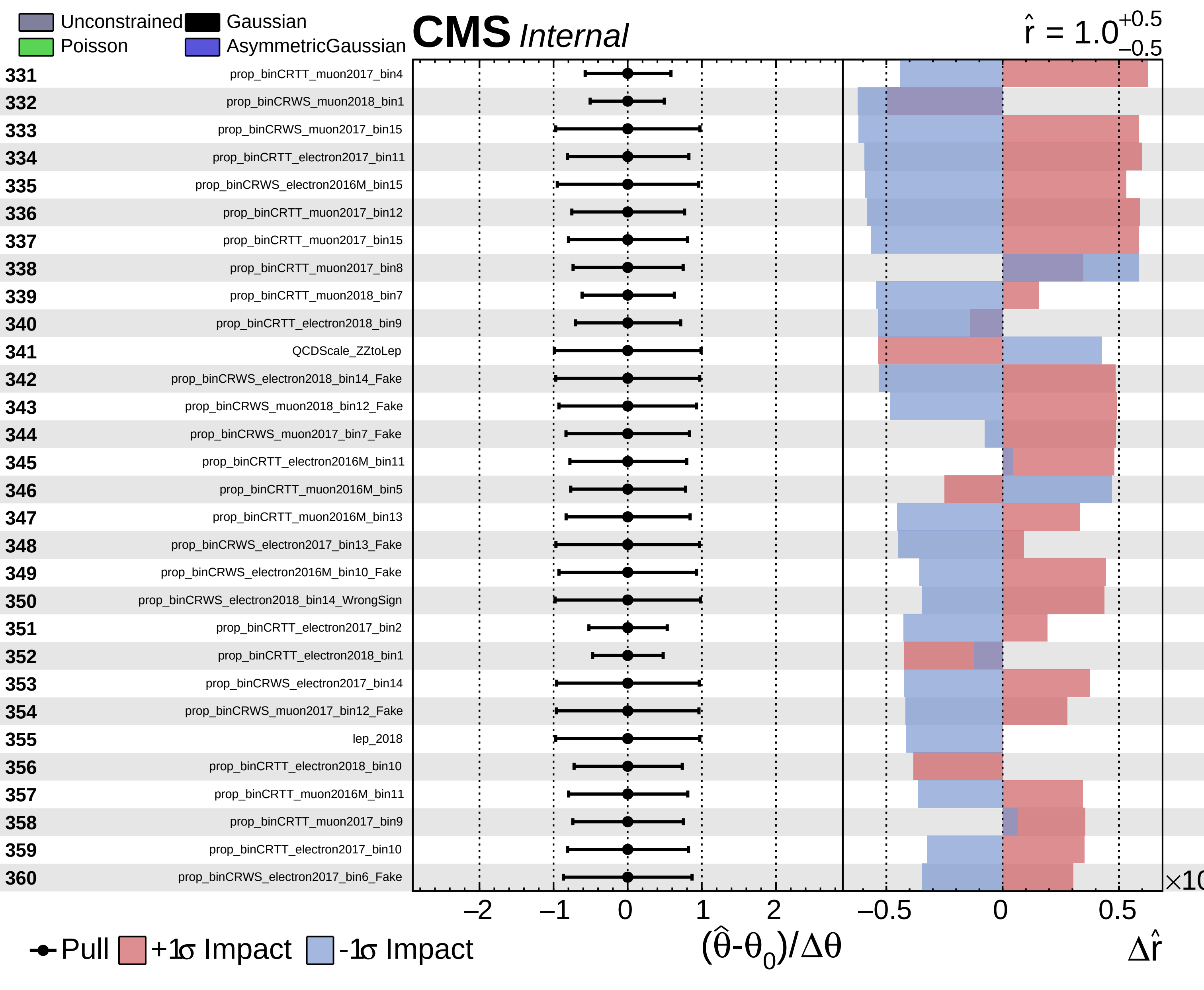
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
241 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
242 prop_binCRWS_muon2018_bin4
243 QCDScale_WrongSign
244 prop_binSR_muon2017_bin0
245 prop_binCRTT_electron2016M_bin1
246 prop_binCRTT_muon2017_bin16
247 prop_binCRTT_muon2018_bin15
248 prop_binCRWS_electron2018_bin12
249 prop_binCRWS_muon2016M_bin13
250 prop_binCRTT_muon2018_bin5
251 prop_binSR_muon2017_bin12_ZZtoLep
252 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
253 prop_binCRTT_muon2018_bin13
254 prop_binCRTT_muon2017_bin14
255 prop_binCRTT_muon2016M_bin10
256 prop_binCRWS_electron2016M_bin7
257 prop_binCRTT_muon2016M_bin2
258 prop_binCRWS_electron2016M_bin14
259 prop_binCRWS_muon2016M_bin12
260 prop_binSR_electron2017_bin11_ZZtoLep
261 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
262 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
263 prop_binCRWS_electron2016M_bin13
264 prop_binCRTT_electron2018_bin15
265 prop_binCRWS_electron2017_bin3
266 prop_binCRTT_electron2017_bin15
267 prop_binCRTT_electron2017_bin16
268 prop_binSR_muon2017_bin12_WpWpJJ
269 prop_binCRWS_muon2018_bin6
270 prop_binCRWS_electron2018_bin6_Fake
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



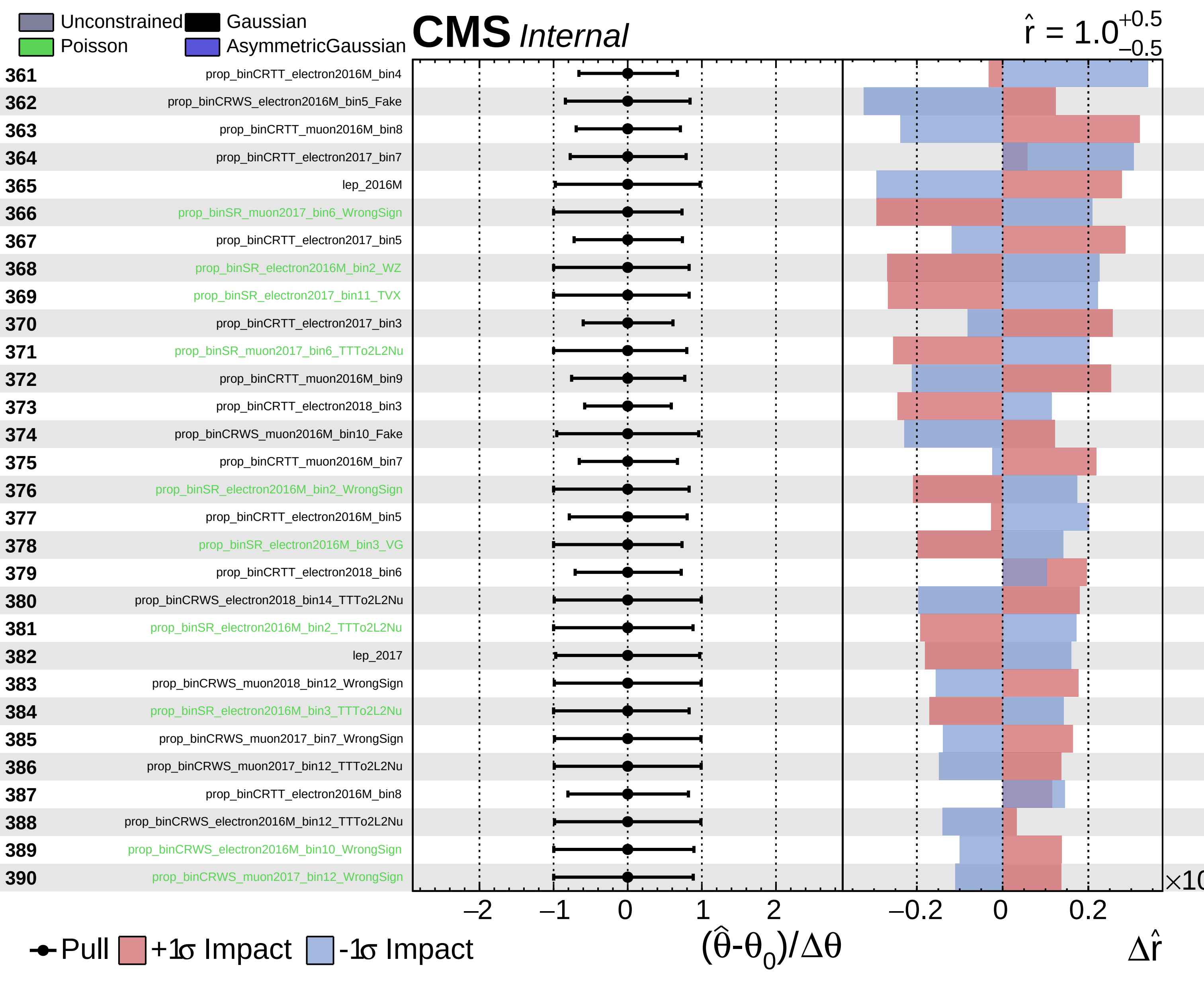
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
271 prop_binCRWS_electron2018_bin8_Fake
272 prop_binCRTT_muon2016M_bin15
273 prop_binCRTT_electron2018_bin14
274 prop_binCRTT_muon2016M_bin16
275 prop_binCRTT_electron2018_bin11
276 prop_binSR_electron2018_bin0
277 prop_binCRTT_electron2017_bin12
278 prop_binCRTT_electron2018_bin8
279 prop_binSR_muon2017_bin12_WrongSign
280 prop_binCRTT_muon2017_bin10
281 prop_binCRTT_electron2016M_bin7
282 prop_binCRTT_electron2016M_bin16
283 prop_binCRTT_electron2017_bin4
284 prop_binCRTT_electron2017_bin14
285 prop_binCRTT_muon2018_bin9
286 prop_binCRWS_electron2016M_bin11
287 prop_binCRTT_muon2016M_bin3
288 prop_binCRTT_electron2018_bin13
289 prop_binCRWS_electron2016M_bin4
290 puID
291 prop_binSR_electron2017_bin11_WrongSign
292 prop_binCRTT_muon2018_bin4
293 prop_binCRTT_electron2016M_bin3
294 prop_binCRWS_muon2017_bin13
295 prop_binCRTT_electron2016M_bin2
296 prop_binSR_electron2017_bin0
297 prop_binCRTT_electron2016M_bin13
298 prop_binCRWS_electron2017_bin9
299 prop_binCRTT_electron2017_bin13
300 prop_binCRTT_muon2017_bin6
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



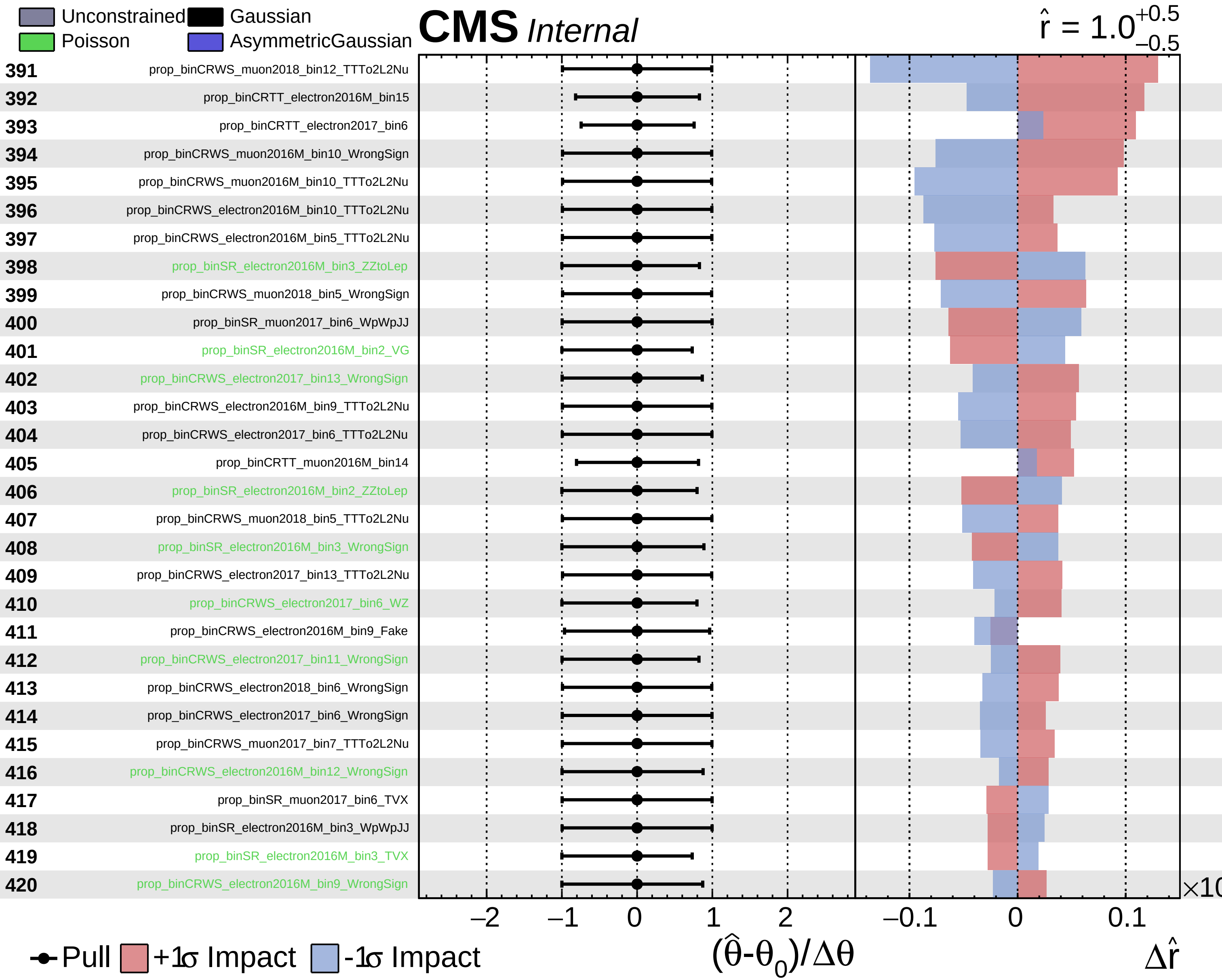
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.5
301 prop_binCRTT_muon2017_bin7
302 prop_binSR_muon2017_bin12_TVX
303 prop_binCRTT_muon2017_bin13
304 prop_binCRWS_electron2016M_bin8
305 prop_binCRTT_muon2017_bin11
306 prop_binCRWS_electron2016M_bin1
307 prop_binCRTT_electron2018_bin7
308 prop_binSR_electron2017_bin11_WpWpJJ
309 prop_binCRTT_electron2016M_bin14
310 prop_binCRWS_muon2016M_bin15
311 prop_binCRTT_muon2016M_bin4
312 prop_binCRTT_muon2016M_bin6
313 prop_binCRWS_electron2018_bin3
314 prop_binCRTT_electron2018_bin4
315 prop_binCRTT_electron2017_bin9
316 prop_binSR_muon2018_bin0
317 prop_binCRWS_electron2017_bin11_Fake
318 prop_binCRTT_electron2016M_bin12
319 prop_binSR_electron2017_bin11_TTTo2L2Nu
320 prop_binCRTT_electron2016M_bin9
321 prop_binCRTT_electron2017_bin8
322 prop_binCRWS_muon2018_bin5_Fake
323 prop_binCRTT_muon2016M_bin12
324 prop_binCRTT_electron2018_bin12
325 prop_binCRTT_electron2016M_bin6
326 prop_binCRTT_electron2018_bin5
327 prop_binCRWS_muon2016M_bin14
328 prop_binSR_electron2016M_bin0
329 prop_binCRWS_electron2018_bin11
330 prop_binCRTT_electron2016M_bin10
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

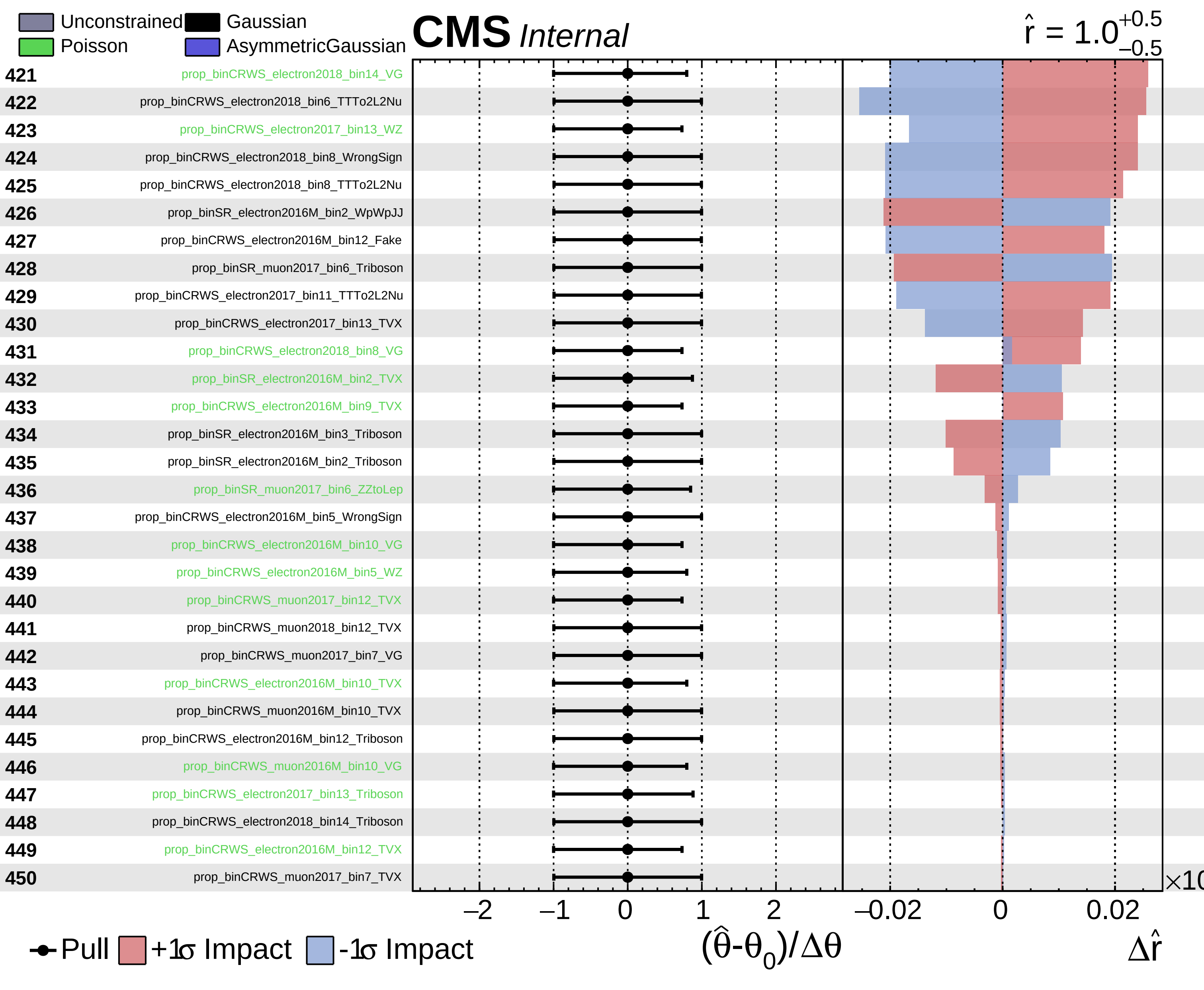


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
331 prop_binCRTT_muon2017_bin4
332 prop_binCRWS_muon2018_bin1
333 prop_binCRWS_muon2017_bin15
334 prop_binCRTT_electron2017_bin11
335 prop_binCRWS_electron2016M_bin15
336 prop_binCRTT_muon2017_bin12
337 prop_binCRTT_muon2017_bin15
338 prop_binCRTT_muon2017_bin8
339 prop_binCRTT_muon2018_bin7
340 prop_binCRTT_electron2018_bin9
341 QCDScale_ZZtoLep
342 prop_binCRWS_electron2018_bin14_Fake
343 prop_binCRWS_muon2018_bin12_Fake
344 prop_binCRWS_muon2017_bin7_Fake
345 prop_binCRTT_electron2016M_bin11
346 prop_binCRTT_muon2016M_bin5
347 prop_binCRTT_muon2016M_bin13
348 prop_binCRWS_electron2017_bin13_Fake
349 prop_binCRWS_electron2016M_bin10_Fake
350 prop_binCRWS_electron2018_bin14_WrongSign
351 prop_binCRTT_electron2017_bin2
352 prop_binCRTT_electron2018_bin1
353 prop_binCRWS_electron2017_bin14
354 prop_binCRWS_muon2017_bin12_Fake
355 lep_2018
356 prop_binCRTT_electron2018_bin10
357 prop_binCRTT_muon2016M_bin11
358 prop_binCRTT_muon2017_bin9
359 prop_binCRTT_electron2017_bin10
360 prop_binCRWS_electron2017_bin6_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

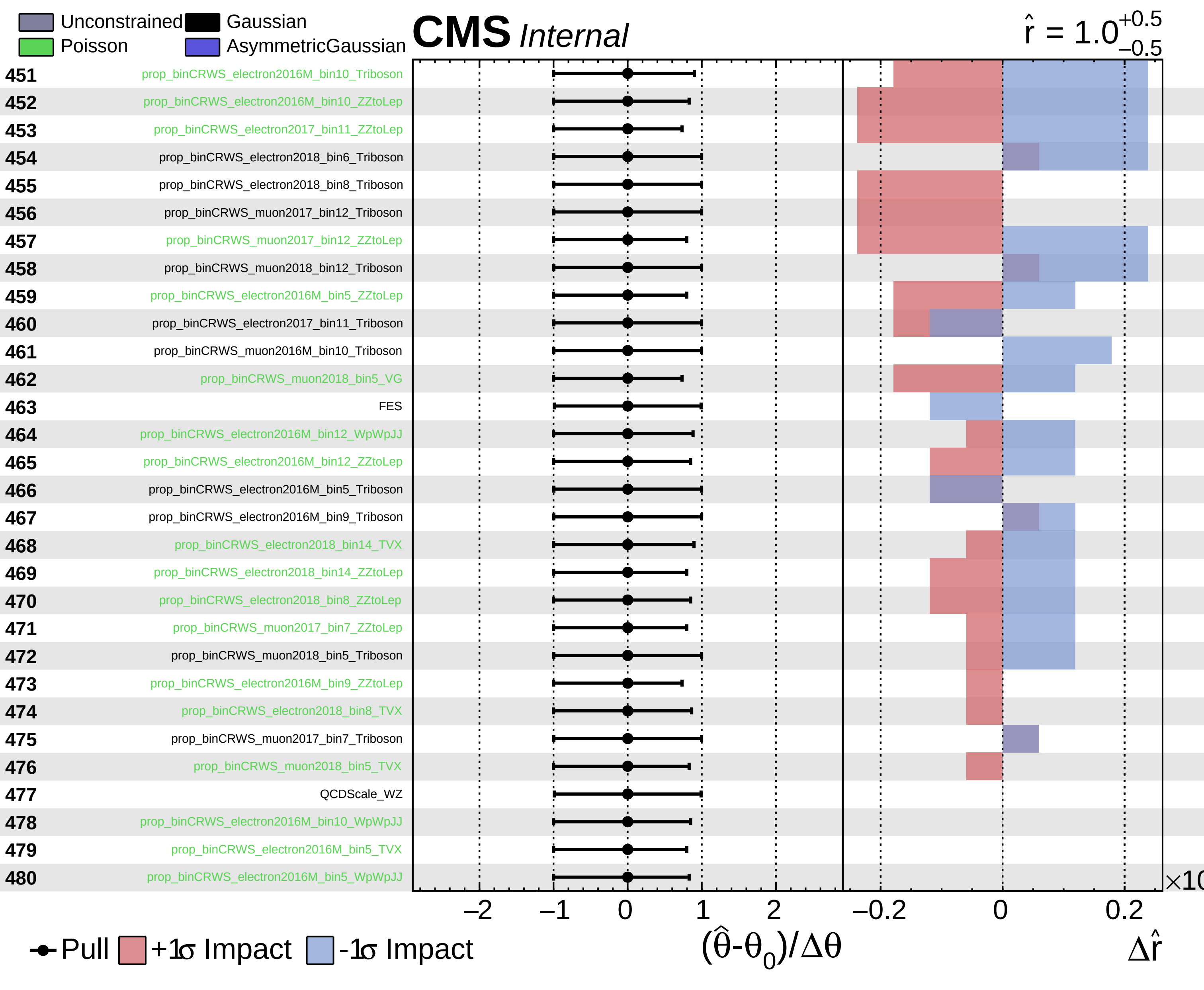


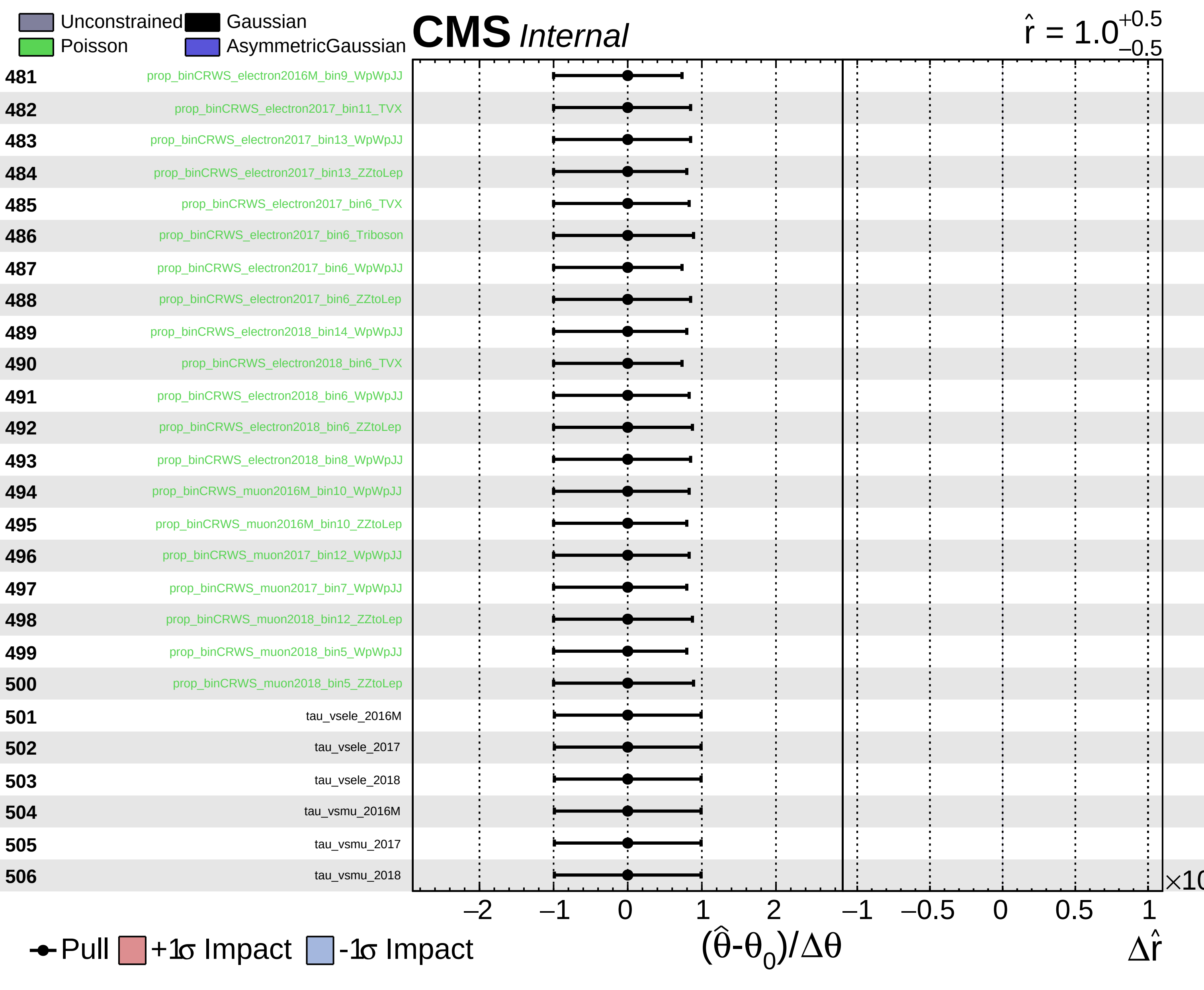
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
361 prop_binCRTT_electron2016M_bin4
362 prop_binCRWS_electron2016M_bin5_Fake
363 prop_binCRTT_muon2016M_bin8
364 prop_binCRTT_electron2017_bin7
365 lep_2016M
366 prop_binSR_muon2017_bin6_WrongSign
367 prop_binCRTT_electron2017_bin5
368 prop_binSR_electron2016M_bin2_WZ
369 prop_binSR_electron2017_bin11_TVX
370 prop_binCRTT_electron2017_bin3
371 prop_binSR_muon2017_bin6_TTTo2L2Nu
372 prop_binCRTT_muon2016M_bin9
373 prop_binCRTT_electron2018_bin3
374 prop_binCRWS_muon2016M_bin10_Fake
375 prop_binCRTT_muon2016M_bin7
376 prop_binSR_electron2016M_bin2_WrongSign
377 prop_binCRTT_electron2016M_bin5
378 prop_binSR_electron2016M_bin3_VG
379 prop_binCRTT_electron2018_bin6
380 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
381 prop_binSR_electron2016M_bin2_TTTo2L2Nu
382 lep_2017
383 prop_binCRWS_muon2018_bin12_WrongSign
384 prop_binSR_electron2016M_bin3_TTTo2L2Nu
385 prop_binCRWS_muon2017_bin7_WrongSign
386 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
387 prop_binCRTT_electron2016M_bin8
388 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
389 prop_binCRWS_electron2016M_bin10_WrongSign
390 prop_binCRWS_muon2017_bin12_WrongSign
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
391 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
392 prop_binCRTT_electron2016M_bin15
393 prop_binCRTT_electron2017_bin6
394 prop_binCRWS_muon2016M_bin10_WrongSign
395 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
396 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
397 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
398 prop_binSR_electron2016M_bin3_ZZtoLep
399 prop_binCRWS_muon2018_bin5_WrongSign
400 prop_binSR_muon2017_bin6_WpWpJJ
401 prop_binSR_electron2016M_bin2_VG
402 prop_binCRWS_electron2017_bin13_WrongSign
403 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
404 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
405 prop_binCRTT_muon2016M_bin14
406 prop_binSR_electron2016M_bin2_ZZtoLep
407 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
408 prop_binSR_electron2016M_bin3_WrongSign
409 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
410 prop_binCRWS_electron2017_bin6_WZ
411 prop_binCRWS_electron2016M_bin9_Fake
412 prop_binCRWS_electron2017_bin11_WrongSign
413 prop_binCRWS_electron2018_bin6_WrongSign
414 prop_binCRWS_electron2017_bin6_WrongSign
415 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
416 prop_binCRWS_electron2016M_bin12_WrongSign
417 prop_binSR_muon2017_bin6_TVX
418 prop_binSR_electron2016M_bin3_WpWpJJ
419 prop_binSR_electron2016M_bin3_TVX
420 prop_binCRWS_electron2016M_bin9_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
421 prop_binCRWS_electron2018_bin14_VG
422 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
423 prop_binCRWS_electron2017_bin13_WZ
424 prop_binCRWS_electron2018_bin8_WrongSign
425 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
426 prop_binSR_electron2016M_bin2_WpWpJJ
427 prop_binCRWS_electron2016M_bin12_Fake
428 prop_binSR_muon2017_bin6_Triboson
429 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
430 prop_binCRWS_electron2017_bin13_TVX
431 prop_binCRWS_electron2018_bin8_VG
432 prop_binSR_electron2016M_bin2_TVX
433 prop_binCRWS_electron2016M_bin9_TVX
434 prop_binSR_electron2016M_bin3_Triboson
435 prop_binSR_electron2016M_bin2_Triboson
436 prop_binSR_muon2017_bin6_ZZtoLep
437 prop_binCRWS_electron2016M_bin5_WrongSign
438 prop_binCRWS_electron2016M_bin10_VG
439 prop_binCRWS_electron2016M_bin5_WZ
440 prop_binCRWS_muon2017_bin12_TVX
441 prop_binCRWS_muon2018_bin12_TVX
442 prop_binCRWS_muon2017_bin7_VG
443 prop_binCRWS_electron2016M_bin10_TVX
444 prop_binCRWS_muon2016M_bin10_TVX
445 prop_binCRWS_electron2016M_bin12_Triboson
446 prop_binCRWS_muon2016M_bin10_VG
447 prop_binCRWS_electron2017_bin13_Triboson
448 prop_binCRWS_electron2018_bin14_Triboson
449 prop_binCRWS_electron2016M_bin12_TVX
450 prop_binCRWS_muon2017_bin7_TVX
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.5}$
451 prop_binCRWS_electron2016M_bin10_Triboson
452 prop_binCRWS_electron2016M_bin10_ZZtoLep
453 prop_binCRWS_electron2017_bin11_ZZtoLep
454 prop_binCRWS_electron2018_bin6_Triboson
455 prop_binCRWS_electron2018_bin8_Triboson
456 prop_binCRWS_muon2017_bin12_Triboson
457 prop_binCRWS_muon2017_bin12_ZZtoLep
458 prop_binCRWS_muon2018_bin12_Triboson
459 prop_binCRWS_electron2016M_bin5_ZZtoLep
460 prop_binCRWS_electron2017_bin11_Triboson
461 prop_binCRWS_muon2016M_bin10_Triboson
462 prop_binCRWS_muon2018_bin5_VG
463 FES
464 prop_binCRWS_electron2016M_bin12_WpWpJJ
465 prop_binCRWS_electron2016M_bin12_ZZtoLep
466 prop_binCRWS_electron2016M_bin5_Triboson
467 prop_binCRWS_electron2016M_bin9_Triboson
468 prop_binCRWS_electron2018_bin14_TVX
469 prop_binCRWS_electron2018_bin14_ZZtoLep
470 prop_binCRWS_electron2018_bin8_ZZtoLep
471 prop_binCRWS_muon2017_bin7_ZZtoLep
472 prop_binCRWS_muon2018_bin5_Triboson
473 prop_binCRWS_electron2016M_bin9_ZZtoLep
474 prop_binCRWS_electron2018_bin8_TVX
475 prop_binCRWS_muon2017_bin7_Triboson
476 prop_binCRWS_muon2018_bin5_TVX
477 QCDScale_WZ
478 prop_binCRWS_electron2016M_bin10_WpWpJJ
479 prop_binCRWS_electron2016M_bin5_TVX
480 prop_binCRWS_electron2016M_bin5_WpWpJJ
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.5
481 prop_binCRWS_electron2016M_bin9_WpWpJJ
482 prop_binCRWS_electron2017_bin11_TVX
483 prop_binCRWS_electron2017_bin13_WpWpJJ
484 prop_binCRWS_electron2017_bin13_ZZtoLep
485 prop_binCRWS_electron2017_bin6_TVX
486 prop_binCRWS_electron2017_bin6_Triboson
487 prop_binCRWS_electron2017_bin6_WpWpJJ
488 prop_binCRWS_electron2017_bin6_ZZtoLep
489 prop_binCRWS_electron2018_bin14_WpWpJJ
490 prop_binCRWS_electron2018_bin6_TVX
491 prop_binCRWS_electron2018_bin6_WpWpJJ
492 prop_binCRWS_electron2018_bin6_ZZtoLep
493 prop_binCRWS_electron2018_bin8_WpWpJJ
494 prop_binCRWS_muon2016M_bin10_WpWpJJ
495 prop_binCRWS_muon2016M_bin10_ZZtoLep
496 prop_binCRWS_muon2017_bin12_WpWpJJ
497 prop_binCRWS_muon2017_bin7_WpWpJJ
498 prop_binCRWS_muon2018_bin12_ZZtoLep
499 prop_binCRWS_muon2018_bin5_WpWpJJ
500 prop_binCRWS_muon2018_bin5_ZZtoLep
501 tau_vsele_2016M
502 tau_vsele_2017
503 tau_vsele_2018
504 tau_vsmu_2016M
505 tau_vsmu_2017
506 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂